

Fig. 2. (Left) The carbon intensity of each resources. We exclude biogas, biomass, geothermal, and coal since it has relatively low carbon contribution in California. (Right) Daily average power supply per resource. We exclude coal and hydro since it is less significant.

Carbon intensity depends on the power grid's operational plan and carbon intensity of each energy resource. In this paper, we specifically examined the carbon intensity for the California grid with data provided by CAISO (California Independent System Operator)². To calculate the carbon intensity, we consulted the carbon emission of the grid resources as well as the supply trend of the grid. Figure 2 describes the feature of the two data sets. In Fig 2 (left) there are two main contributors to the state's grid emission: imports and natural gas.

$$C(t) = \sum_{i=1}^N C_i(t) \cdot \frac{P_i(t)}{P_{total}(t)}, \forall i \in \text{grid mix}; \quad (1)$$

where $C_i(t)$ is a time-series data of the carbon intensity of each energy source, $P_i(t)$ is the power supplied by the energy source and $P_{total}(t)$ is the total power generated on the specific day at time t . From the formula and the data, we can expect the carbon intensity to be high during the night and the early morning when solar energy is less available. On the other hand, the carbon intensity of the grid tends to be low during the day as solar energy and other renewables become a major contributor to the state's power supply.

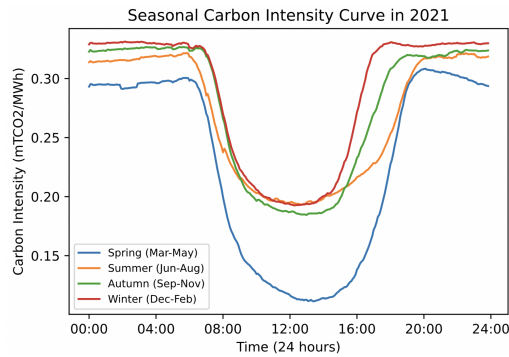


Fig. 3. Average carbon intensity during different seasons.

Carbon intensity is not only grid-mix dependent and time sensitive, but is also seasonally sensitive. Fig 3 shows an

example of carbon intensity calculated using the above formula throughout the year. We notice that carbon intensity is lower especially during the spring. On top of that, we also notice a shift in time for the carbon intensity curve in autumn as well as winter. We can attribute this to the daylight saving hours policy in California.

B. Charging Station Operation Model

As mentioned in Section II, our charging framework takes into account of the user constraints as well as the infrastructure constraints. These constraints are discussed below. We use the discrete time model, where the time step is indexed in $\mathcal{T} := \{1, 2, 3, \dots, T\}$. $\mathcal{V}$ is the set of all EVs. For each vehicle $i \in \mathcal{V}$, we denote the charging demand information is $(x_{i,arrival}, x_{i,depart}, t_{i,arrival}, t_{i,depart})$, where $x_{i,arrival}$ is the state-of-charge SoC of EV i upon its arrival of the charging station, $x_{i,depart}$ is the desired state-of-charge SOC of EV i upon its departure. and $t_{i,arrival}, t_{i,depart}$ is the time index when the vehicle i arrive in / depart from the charging station, $t_{i,arrival}, t_{i,depart} \in \mathcal{T}$. The above four factors are all externally determined by the user. Arrival time $t_{i,arrival}$ and initial energy level $x_{i,arrival}$ are determined when user plugged in the EV. Departure time $t_{i,depart}$ and charge $x_{i,depart}$ are input by the user when they initiate the charging session.

The model aims at deciding the charging power $u_i(t)$ for vehicle i at time t , and the state-of-charge (SoC) $x_i(t)$ for EV i at time t given the specific objective. We are interested in solving the following optimization problem:

$$\min_{\mathbf{u}} \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{V}} C(t) u_i(t) \Delta T + \lambda \sum_{i \in \mathcal{V}} |x_i(T) - x_{i,depart}| \quad (2a)$$

$$s.t. \quad x_i(0) = x_{i,arrival}, \quad (2b)$$

$$x_i(t) = x_i(t-1) + \frac{\delta u_i(t)}{E_i}, \quad t \geq 1; \quad (2c)$$

$$0 \leq x_i(t) \leq \bar{x}_i, \quad i \in \mathcal{V}; \quad (2d)$$

$$0 \leq u_i(t) \leq \bar{u}_i, \quad i \in \mathcal{V}; \quad (2e)$$

$$\sum_{i \in \mathcal{V}} u_i(t) \leq \bar{P}, \quad t \in \mathcal{T}; \quad (2f)$$

$$u_i(t) = 0, \quad t \notin [t_{i,arrival}, t_{i,depart}), \quad i \in \mathcal{V}. \quad (2g)$$

The above optimal charging scheduling problem minimizes the total carbon emissions while satisfying charging sessions' time and energy constraints, defined by the objective function (2a). We firstly consider the case when $C(t)$ and EV charging demand information for next T steps are available and communicated to the charging station operator. We denote ΔT to be the discretization interval, and in our case, ΔT is 5 minutes aligned with the carbon intensity measurement interval published by CAISO. λ is a balanced factor to trade off between carbon reduction and energy delivery. The operational constraints considered in this framework are described as follows:

EV charging constraints: Constraint (2b) defines the SoC level of EV upon arrival; and (2c) depicts the dynamics between charging power and battery SoC, where E_i is the battery

²<http://www.caiso.com/todaysoutlook/pages/emissions.html>

energy capacity (kWh). δ is an EV-specific charging parameter, which is known before the operation and is computed based on the charging efficiency and discretization, e.g., if charging efficiency is 90%, and discretization interval is 15 minutes, $\delta = 0.9 \times 15/60 = 0.225$. (2d) and (2e) constrain the admissible charging power for single EV, where the battery maximum SoC is $\bar{x}_i$ and maximum charging power is $\bar{u}_i$.

Charging station constraints: (2f) limits the total available charging power $\bar{P}$ at the charging station. We note that in this paper, we assume there is a known $\bar{P}$ decided by the charging station's transformer.

Temporal logic of charging sessions: (2g) describes the temporal logic that the charging session has to satisfy, as no charging power shall be delivered whenever EV i has not arrived ($t < t_{i,arrival}$) or has departed ($t \geq t_{i,depart}$).

Other Factors to generalize: The optimization problem that we introduced in (2a) is a general model that can be further generalized. For example, battery health is another important factor to be considered in EV charging. A more complex version of this optimization problem could potentially include penalty terms in the objective to discourage sudden change in charging power, which prevents battery health from deteriorating.

IV. REAL-TIME SCHEDULING

In this section, we propose a forecasting module to inform charging station operators day-ahead carbon intensity, and extend the framework described in Sec III to real-time settings.

Offline algorithm assumes we have access to all the information including future arrival statistics. While in the real-time algorithm, we assume we can only decide charging rate based on the predicted carbon intensity and information of arrived electric vehicles. An online algorithm is crucial for the real life applicability of our charging strategy because there are two main factors that changes daily in our optimization problem: daily average carbon intensity and daily EV arrival pattern. From figure 5 and 6, we can see that daily EV arrival varies from 5 vehicles to more than 40 vehicles per day. An online scheduling algorithm will be able to adjust scheduling the given arrival of EV's. On top of that, our discussion of carbon intensity also demonstrated seasonal volatility. With EV arrival uncertainty combined with carbon intensity uncertainty, and online algorithm allows the charging station to adjust its charging strategy given different scenario.

A. Carbon Intensity Day-Ahead Forecasting

We use the 2021 CAISO data (shown in Fig 2) and CAISO day-ahead load forecast data $Load(t)$ to fit and evaluate the carbon predictor model. The grid's day-ahead carbon intensity $\hat{C}(t)$ is closely related to the future electricity demand, calendar features, and past carbon intensity patterns. We randomly choose 80% data to fit the linear regression model and rest data for testing the model performance. The resulting forecasting model

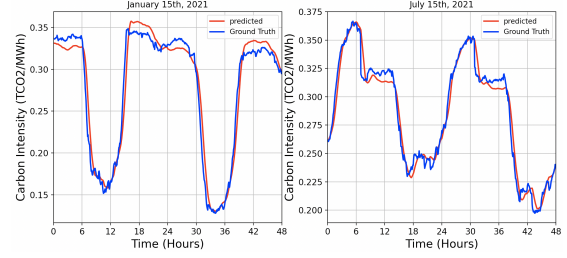


Fig. 4. The ground truth (blue) and predicted carbon intensity (red) for two-day sample in January and July respectively.

is as follows:

$$\begin{aligned} \hat{C}(t) = & \beta_0 + \beta_1 \times Minute + \beta_2 \times Hour + \beta_3 \times Day \\ & + \beta_4 \times Month + \beta_5 \times Load(t) + \beta_6 \times MA(C(t))_{24} \\ & + \beta_7 \times MA(C(t))_{12} + \beta_8 \times MA(C(t))_1; \end{aligned} \quad (3)$$

where $MA(\cdot)_a$ denotes the moving average of past a hours. Fig 4 illustrates the ground truth carbon intensity and forecasted carbon intensity using the fitted model (3). We note that other statistical and machine learning models can also be used for carbon intensity prediction. We use the linear regression model since it is simple and provides accurate prediction result. For the CAISO case study, the overall testing mean squared absolute error is 0.0085. We also demonstrate such forecasts can be readily integrated into the proposed charging scheduler without affecting the algorithm performance. Power grid CO2 emission intensity forecast is also investigated in [21] by considering additional features and an autoregressive model. We calculate and forecast the average carbon intensity rather than marginal carbon intensity, as in this study, we focus on the operating strategies for single EV charging station. Such single-station operation scheme shall pose minimal effects to transmission grid carbon profiles. We refer to [22] for a more complete review and open-source data on electricity grid carbon intensities.

B. Online Algorithm

With the predicted carbon intensity $\hat{C}(t)$, we are ready to implement proposed carbon-aware charging scheduler in an online fashion. Denote the optimization problem (2) as **SCH** with input information coming from the charging infrastructure and EVs' peak power constraints $\bar{P}$ and $\bar{u}$, predicted carbon intensity, the active charging vehicles $\mathcal{V}_k$ along with their states $x_{\mathcal{V}_k}$ where $k \in \mathcal{K}, \mathcal{K} := \{1, 2, 3, \dots, K\}$, the algorithm is summarized in Algorithm 1.

At each time step $k = 1, 2, \dots, K$, we take the forecasted carbon intensity for the next T steps, $\mathcal{C}_k := \{\hat{C}(k), \dots, \hat{C}(k+T-1)\}$, the current available EVs $\mathcal{V}_k$ and their SoC levels $x_{\mathcal{V}_k}$, and solves the optimization (2). With the optimized charging schedule $(u_i^*(1), \dots, u_i^*(T), i \in \mathcal{V}_k)$, we assign $u_i^*(1)$ to the charging power of EV i at time step k to $u_i(k) = u_i^*(1)$ for the active vehicles and update state of charge accordingly for every vehicle $x_i(k+1) = x_i(k) + \frac{\delta u_i(k)}{E_i}, \forall i \in \mathcal{V}$.